

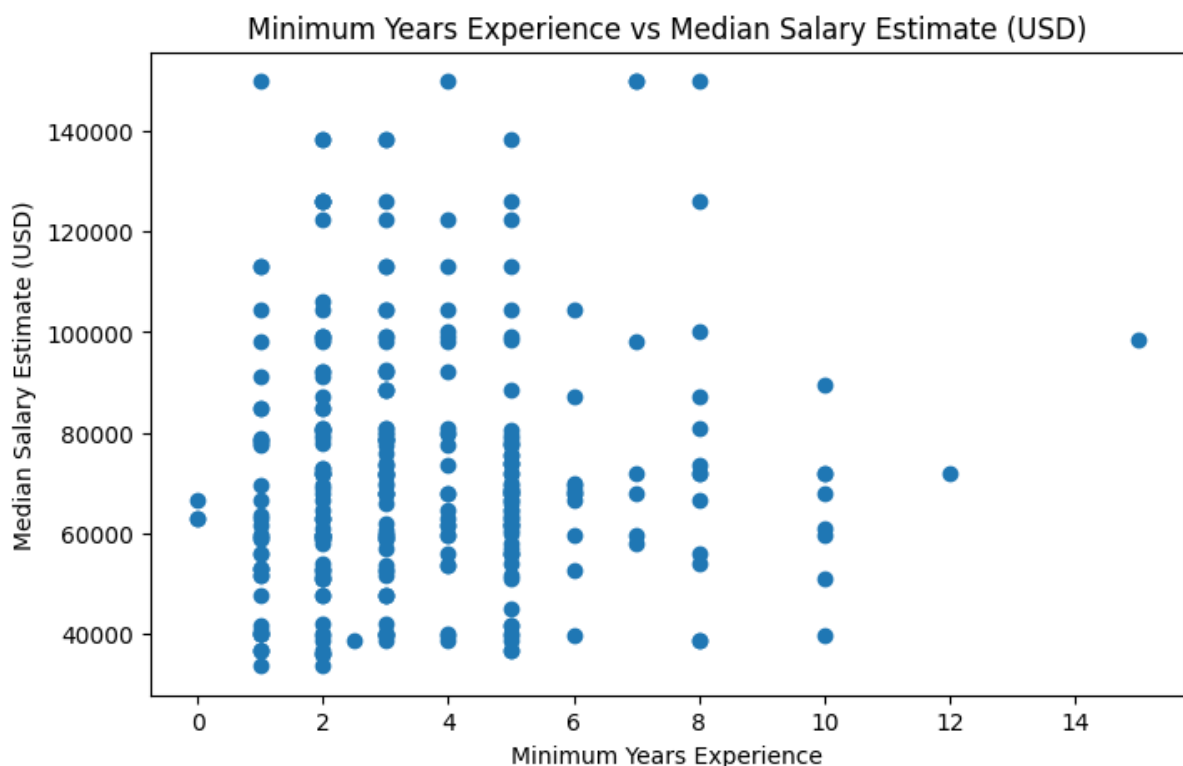
Analyzing Job Market Data with LLMs in Sheets and AI Reporting Tools

Introduction

This report analyzes the relationship between job requirements and salary estimates in data analyst job postings. Using data visualizations, the report explores how minimum years of experience and programming language requirements relate to median salary estimates.

1) Years of Experience Analysis

Scatter plot: Minimum Years Experience vs. Median Salary Estimate

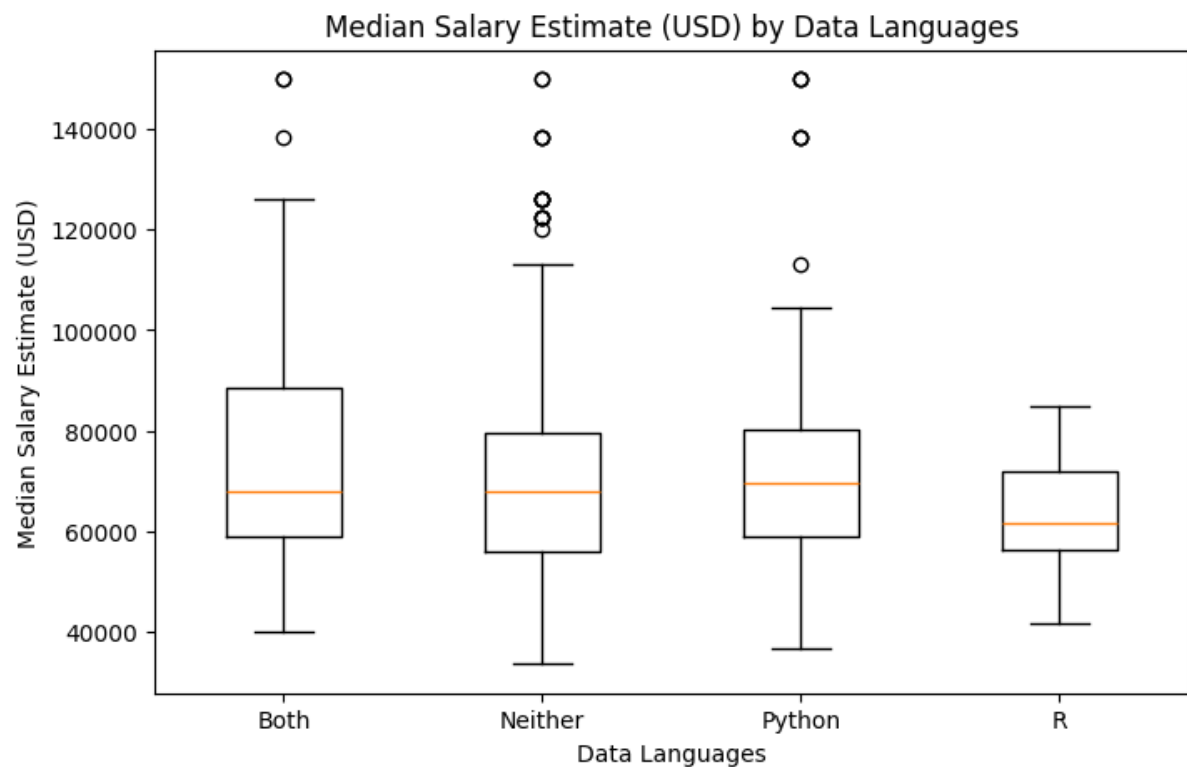


The scatter plot shows a weak positive relationship between years of experience and salary. Salaries generally increase slightly with experience, but there is a lot of variation between jobs.

In this sample, the correlation is close to zero, which means salary does not increase steadily as experience requirements rise. Instead, the points are widely spread, showing that jobs with similar experience requirements can have very different salary estimates.

2) Programming Language Analysis

Box plot: Median Salary Estimate grouped by Data Languages



The box plot suggests that jobs requiring Python or both Python and R tend to have higher salaries than jobs requiring neither language. However, salary ranges overlap across groups.

The largest salary spread appears in the Python and Both categories, which indicates that some positions pay much more than others even within the same language group. This may reflect differences in seniority, industry, or job scope. It is also important to note that the R-only group is small, so its results should be interpreted cautiously.

3) Reflection

Creating the visualizations was straightforward after cleaning the data, but interpreting weak relationships was more difficult. I learned that salary is influenced by multiple factors, and data visualization helps reveal patterns that are not obvious in tables alone.